

Notes: Ferraz and Finan (QJE, 2008)
**Exposing Corrupt Politicians: The Effects of Brazil’s Publicly Released Audits
on Electoral Outcomes**

Contents

1	Big picture	2
2	Incremental contribution of the paper	2
2.1	Contribution relative to political accountability theory	2
2.2	Contribution relative to corruption research	2
2.3	Contribution relative to media and information diffusion	3
2.4	Contribution relative to randomized policy evaluation	3
3	Institutional background and data	3
3.1	Brazil’s public audit program	3
3.2	Treatment and control groups	3
3.3	Corruption measure	4
3.4	Electoral outcomes	4
3.5	Media data	4
4	Empirical specifications	4
4.1	Average audit effect	4
4.2	Audit effects by corruption level	5
4.3	Flexible corruption specifications	5
4.4	Media interactions	5
5	Identification and threats	5
5.1	Randomized timing	5
5.2	Potential threat: audit manipulation	6
5.3	Potential threat: endogenous corruption and media	6
6	Tables and figures	6
6.1	Table I and Figures I–II: balance, corruption distribution, and timing	6
6.2	Table II and Figure III: average effects and corruption-gradient evidence	7
6.3	Tables III and IV: regression evidence across electoral outcomes	8
6.4	Table V: audit quality and robustness checks	9
6.5	Table VI and Figure IV: local radio and dissemination	9
7	Short summary	10
8	Conclusion	11
8.1	Core conclusion	11
8.2	Economic intuition	11
8.3	Takeaway	11

1 Big picture

- **Question.** Does disclosing credible information about local government corruption change voter behavior and improve electoral accountability?
- **Setting.** Brazil's federal government randomly audited municipalities for the use of federally transferred funds beginning in 2003. Audit reports were publicly released and disseminated to prosecutors and media outlets.
- **Main empirical comparison.** The paper compares municipalities audited before the 2004 municipal election with municipalities audited after the election, conditioning on the same realized level of corruption reported in the audit.
- **Core outcome.** The main dependent variable is whether the incumbent mayor was reelected in 2004, with additional outcomes such as vote share, win margin, and changes relative to the 2000 election.
- **Main result.** Publicly released audits reduced reelection prospects for corrupt incumbents. When two corrupt violations were revealed, reelection probability fell by about seven percentage points; with three corrupt violations, the effect roughly doubled.
- **Media result.** Effects are larger when a local radio station is present. Radio makes bad news more politically damaging and good news more politically beneficial.
- **Key interpretation.** Voters update beliefs about incumbent quality when credible audit information is released. The audit program therefore strengthens political selection.
- **Broader lesson.** Transparency policies matter most when information is credible, timely, and disseminated through media channels that voters actually use.

2 Incremental contribution of the paper

2.1 Contribution relative to political accountability theory

- Theoretical models of elections emphasize that voters can discipline politicians only when they observe enough information about incumbent performance.
- Before this paper, direct empirical evidence was limited because information about politician quality is usually endogenous: voters, journalists, opposition parties, and prosecutors uncover information selectively.
- Ferraz and Finan exploit a government audit program in which municipalities are randomly selected and audit reports are publicly released.

Interpretation: The paper provides unusually clean evidence that better information about incumbent malfeasance changes election outcomes. This turns a central theoretical claim into an empirically identified result.

2.2 Contribution relative to corruption research

- Existing work often documents corruption or estimates its economic costs, but it is harder to show whether voters punish corruption once they learn about it.
- This paper constructs an objective corruption measure directly from audit reports, classifying corruption violations such as fraud, diversion of funds, overinvoicing, or irregular procurement.
- It shows that voters punish corruption when information is released before the election, and that the punishment is increasing in the number of reported corrupt violations.

Interpretation: The paper does not merely show that corrupt politicians are unpopular. It shows that a randomized information shock about corruption causes electoral punishment.

2.3 Contribution relative to media and information diffusion

- The paper jointly studies transparency and media access. A public audit report may not matter if voters do not hear about it.
- Local radio strongly amplifies the electoral consequences of the audits.
- The same media channel rewards clean incumbents: in municipalities with radio and no corruption, reelection probabilities rise.

Interpretation: The paper's incremental contribution is to show that information disclosure and local media are complements. Transparency requires dissemination infrastructure.

2.4 Contribution relative to randomized policy evaluation

- The audit program is not a standard researcher-designed RCT, but it creates randomized timing of public information.
- The treatment is not the audit itself; the treatment is release of audit information before the election.
- The design is close to an ideal experiment: compare electoral outcomes for identical corruption levels when information is released before versus after voters choose.

Interpretation: The paper is an early and influential example of using government administrative randomization to study political economy mechanisms.

3 Institutional background and data

3.1 Brazil's public audit program

- In 2003, Brazil's federal anticorruption agency, the Controladoria Geral da Uniao (CGU), began randomly selecting municipalities for audits.
- Auditors examined the use of federally transferred funds across programs such as health, education, infrastructure, and social assistance.
- After the audits, reports were released publicly, sent to prosecutors, and made available to media outlets.
- Municipalities were selected by lottery, making selection unrelated to unobserved political trends.

Interpretation: Random selection solves a core identification problem: corrupt or politically vulnerable municipalities are not more likely to be audited because of their own characteristics.

3.2 Treatment and control groups

- The treatment group consists of municipalities audited before the October 2004 municipal election.
- The control group consists of municipalities audited after the 2004 election.
- The main sample focuses on municipalities whose mayors were eligible to run for reelection in 2004.
- Because audits after the election reveal corruption that voters could not yet observe, the postelection group provides a valid control for the corruption distribution that would have been reported under treatment.

Interpretation: The key treatment is not corruption. Corruption is observed in both groups. The treatment is whether the audit information was released before voters made their decision.

3.3 Corruption measure

- The authors read the audit reports and code the number of violations associated with corruption.
- Examples include diversion of funds, procurement fraud, overinvoicing, irregular public works contracts, and missing goods or services.
- The main explanatory variable is the number of corrupt violations.
- The analysis also uses flexible nonlinear functions of corruption because electoral punishment need not be linear in the number of violations.

Interpretation: A crucial strength of the paper is that corruption is measured by independent auditors rather than by media reports, accusations, or political opponents.

3.4 Electoral outcomes

- The main outcome is whether the incumbent mayor was reelected in 2004.
- Other outcomes include incumbent vote share, win margin, and changes in vote share and win margin relative to the 2000 election.
- The paper also controls for state fixed effects and mayoral and municipal characteristics.

Interpretation: Reelection is the most direct political accountability measure, but vote shares and margins show whether the audit information changed electoral support more continuously.

3.5 Media data

- Local radio station presence is measured at the municipality level.
- The paper also considers the number of radio stations and household radio ownership as alternative media-access measures.
- Radio is the most relevant medium in many small Brazilian municipalities because local newspapers and television are less prevalent.

Interpretation: The media variable tests whether information disclosure reaches voters. Public information only matters if it is disseminated and credible.

4 Empirical specifications

4.1 Average audit effect

The basic reduced-form specification is:

$$E_{ms} = \alpha + \beta A_{ms} + X'_{ms} \gamma + \nu_s + \varepsilon_{ms}, \quad (1)$$

where:

- E_{ms} is an electoral outcome for municipality m in state s ;
- A_{ms} equals one if the municipality was audited before the 2004 election;
- X_{ms} contains municipality and mayor characteristics;

- ν_s are state fixed effects.

Interpretation: This regression estimates the average effect of being audited before the election, regardless of whether the audit revealed bad or good news.

4.2 Audit effects by corruption level

The main specification interacts preelection audits with corruption:

$$E_{ms} = \alpha + \beta A_{ms} + \eta C_{ms} + \delta(A_{ms} \times C_{ms}) + X'_{ms}\gamma + \nu_s + \varepsilon_{ms}, \quad (2)$$

where C_{ms} is the number of reported corrupt violations.

- η captures the relationship between corruption and electoral outcomes in the control group.
- δ captures the additional effect of corruption when the audit is released before the election.
- A negative δ means that voters punish corruption when they learn about it.

Interpretation: This is the central specification. Corruption itself is not randomized, but the timing of information release is randomized conditional on corruption.

4.3 Flexible corruption specifications

The paper estimates linear, quadratic, semiparametric, and spline specifications. The flexible models allow electoral punishment to be nonlinear in corruption.

Interpretation: Flexible specifications are important because voters may respond little to one violation but strongly to multiple violations, or because high corruption levels may have threshold effects.

4.4 Media interactions

To test the role of radio, the paper estimates:

$$E_{ms} = \alpha + \beta A_{ms} + \eta C_{ms} + \rho Radio_m + \delta_1(A_{ms} \times C_{ms}) + \delta_2(A_{ms} \times C_{ms} \times Radio_m) + X'_{ms}\gamma + \nu_s + \varepsilon_{ms}. \quad (3)$$

Interpretation: The triple interaction asks whether corruption information has a stronger electoral effect when local radio can disseminate the audit results.

5 Identification and threats

5.1 Randomized timing

- Municipalities were selected by lottery for auditing.
- Some audit reports were released before the 2004 elections; others were released after.
- Conditional on corruption revealed in the report, the timing of release is plausibly exogenous.

Interpretation: This randomized timing is what lets the authors compare mayors whose corruption was revealed to voters with mayors whose corruption was only revealed after voters had already voted.

5.2 Potential threat: audit manipulation

- The design would fail if preelection auditors investigated more aggressively than postelection auditors.
- The paper tests for imbalances in municipality characteristics and audit characteristics.
- It finds no evidence that politically connected mayors received more favorable audits.

Interpretation: The balance evidence supports the interpretation that differences in electoral outcomes are driven by information release rather than biased audit production.

5.3 Potential threat: endogenous corruption and media

- Corruption and radio presence are not randomly assigned.
- The paper addresses this by comparing preelection and postelection audits at the same corruption levels.
- It also tests whether audit effects differ by correlates of corruption or radio, including education, urbanization, population, political competition, newspapers, and television.

Interpretation: The key random variation is timing. The paper uses controls and robustness checks to show that corruption and radio interactions are not simply capturing other municipal characteristics.

6 Tables and figures

6.1 Table I and Figures I–II: balance, corruption distribution, and timing

Read:

- Table I shows that preelection and postelection audit municipalities are broadly balanced on political, mayoral, and socioeconomic characteristics.
- Figure I shows that the distribution of corruption violations is similar for preelection and postelection audits.
- Figure II shows the timing of report releases, with preelection audits released before October 2004 and postelection audits released after.

Economic message: The design is credible because corruption levels and observables are similar across treatment timing groups.

TABLE I
CHARACTERISTICS OF THE MUNICIPALITIES

	Postelection audit (1)	Preelection audit (2)	Difference (3)	Standard error (4)
<i>Panel A: Political characteristics</i>				
Reelection rates for the 2004 elections	0.413	0.395	0.018	0.045
Reelection rates for the 2000 elections	0.423	0.443	-0.020	0.040
2004 reelection rates, among those that ran	0.585	0.559	0.026	0.044
Ran for reelection in 2004	0.707	0.707	-0.001	0.060
Number of parties in 2000	2.881	2.933	-0.052	0.140
Margin of victory in 2000	0.142	0.131	0.012	0.019
Mayor's vote share in 2000	0.529	0.525	0.004	0.013
<i>Panel B: Mayoral characteristics</i>				
Age	47.5	48.0	-0.5	0.9
Years of education	12.2	12.0	0.3	0.3
Male	0.96	0.94	0.02	0.03
Member of PSB	0.083	0.072	0.011	0.044
Member of PT	0.030	0.048	-0.018	0.023
Member of PMDB	0.254	0.172	0.082	0.047
Member of PFL	0.178	0.163	0.015	0.052
Member of PPB	0.030	0.038	-0.009	0.017
Member of PSDB	0.130	0.167	-0.037	0.043
<i>Panel C: Municipal characteristics</i>				
Population density (Persons/km)	0.57	0.73	-0.16	0.33
Literacy rate (%)	0.81	0.80	0.01	0.03
Urban (%)	0.62	0.62	0.00	0.05
Log per capita income	4.72	4.66	0.06	0.15
Income inequality	0.55	0.54	0.00	0.01
Zoning laws	0.29	0.21	0.08	0.07
Economic incentives	0.66	0.58	0.07	0.06
Paved roads	58.99	58.30	0.69	7.74
Size of public employment	0.42	0.43	-0.01	0.02
Municipal guards	0.20	0.21	-0.01	0.07
Small claims court	0.38	0.34	0.04	0.08
Judiciary district	0.59	0.56	0.03	0.07
Number of newspapers	3.58	2.21	1.37	0.79
Share of households that own a radio	0.79	0.77	-0.02	0.02
Municipalities with a radio station	0.31	0.24	0.07	0.06

TABLE I
(CONTINUED)

	Postelection audit (1)	Preelection audit (2)	Difference (3)	Standard error (4)
Number of radio stations, conditional on having one	1.37	1.29	0.08	0.11
Number of corrupt violations	1.952	1.584	0.369	0.357
Total resources audited (R\$)	5,770,189	5,270,001	500,188	1,361,431
Sample size	168	205		

Notes. This table reports the mean political, mayoral, and socioeconomic characteristics of all the municipalities that were audited in the first thirteen lotteries. With the exception of reelection rates for the 2000 election, these statistics were only computed for the 373 municipalities where the mayor was eligible for reelection. The 2000 reelection rates, which include both first- and second-term mayors, were computed for 669 municipalities. Column (1) reports the means for the 168 municipalities that were audited after the elections and constitute our control group. Column (2) reports the mean for the 205 municipalities that were audited before the elections and hence constitute our treatment group. Column (3) reports the difference in means and column (4) presents the standard error of the difference. The political and mayor characteristics presented in Panels A and B were constructed using data from Brazil's electoral commission (Tribunal Superior Eleitoral: <http://www.tse.gov.br/index.html>). The socioeconomic characteristics presented in Panel C were constructed using data from Brazil's statistical bureau (Instituto Brasileiro de Geografia e Estatística: <http://www.ibge.gov.br>). The corruption measure and the amount of resources audited were constructed from the audit reports conducted by Brazil's controller's office (Controladoria Geral da União: <http://www.cgu.gov.br>). Definition of the variables: *Ran for reelection in 2004* is the proportion of eligible (first-term) mayors who ran for reelection in 2004; *Number of parties in 2000* is the average number of political parties that competed in the 2000 elections; *Margin of victory* is the average difference between the winner and the second highest vote share; *PSB, PT, PMB, PFL, PPB, PSDB* are major political parties in Brazil and accounts for approximately 70% of the mayors in 2004; *Urban* is the share of households that live in urban areas; *Log per capita income* is log of the average monthly per capita income of a household; *Income inequality* is the Gini coefficient computed for monthly income; *Zoning law* is an indicator for whether the municipality has zoning laws; *Economic incentives* is an indicator for whether the municipality provides economic incentives to businesses; *Paved roads* is an indicator for whether the municipality has paved roads; *Size of public employment* is the share of the budget in 1999 that was used to pay public employees; *Municipal guards* is an indicator for whether the municipality has its own police force; *Small claims court* is an indicator for whether the municipality has a small claims court; *Judiciary district* is an indicator for whether the municipality has a judiciary district; *Number of corrupt violations* is the sum of violations that are associated with corruption; *Total resources audited* is the amount of funds that was audited by the CGU, expressed in reais.

Table I, balance characteristics

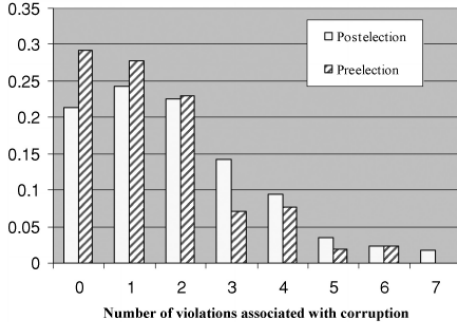


FIGURE I

ribution of Corruption Violations by Pre- versus Postelection Audits

Figure shows the distribution of corruption incidents reported in the he striped bars represent the 168 municipalities that were audited before ons. The solid bars denote 205 municipalities audited after the elections. e was calculated based on our entire sample of municipalities with first- ors, using data from the CGU audit reports.

Figure I: corruption distribution

Table I continued and notes

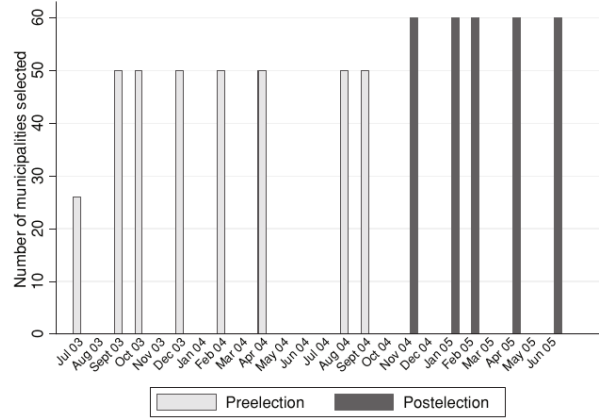


FIGURE II

Timing of the Release of the Audits

Notes. Figure shows the dates for the release of the audit reports for every municipality that was audited in the first thirteen lotteries. The lighter bars denote the 376 municipalities that were audited before the elections. The darker bars denote 300 municipalities audited after the elections. The figure was calculated

Figure II: audit release timing

6.2 Table II and Figure III: average effects and corruption-gradient evidence

Read:

- Table II shows that the average effect of preelection audits is negative but not statistically large when corruption content is ignored.
- Figure III shows why: audit effects differ sharply by the amount of corruption revealed.

- Noncorrupt or lightly corrupt incumbents are not punished, while incumbents with multiple violations are punished strongly.

Economic message: The audit policy is an information treatment, not a uniform anti-incumbent shock. Voters respond to the content of the audit report.

TABLE II
THE AVERAGE EFFECTS OF THE RELEASE OF THE AUDITS ON ELECTORAL OUTCOMES

	All incumbent mayors		Only mayors that ran for reelection				
	Pr(reelection)		Pr(reelection)	Vote share	Win margin	Change in vote share	Change in win margin
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
reelection audit (1/0)	-0.036 [0.053]	-0.036 [0.052]	-0.059 [0.065]	-0.055 [0.072]	-0.090 [0.027]	-0.032 [0.018]+	-0.028 [0.027]
observations	373	373	263	263	263	263	263
fixed effects	0.05	0.17	0.22	0.16	0.22	0.39	0.31
unincorporated effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
ayoral characteristics	No	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table reports the effects of the audits on various electoral outcomes. Each column presents the results of an OLS regression of the dependent variables listed in that row on an indicator variable for whether the municipality was audited before the elections. Except for column (1), all regressions include municipal characteristics: population (persons/km²), percentage of the population that is literate, percentage of the population that lives in the urban sector, per capita income expressed in logarithms, Gini index for income, effective number of political parties in the 2000 mayor election, municipal police (1/0), small claims court (1/0), judiciary district (1/0), municipal characteristics: 0 for male, age, married (1/0), education level, party dummies, and state intercepts. The sample in columns (1) and (2) includes all mayors who were eligible for reelection. The sample in columns (3)–(7) include only the mayors who chose to run for reelection. Robust standard errors are displayed in brackets. Significantly different from zero at 99 (**), 95 (*), and 10 (+) % confidence.

Table II: average effects of released audits

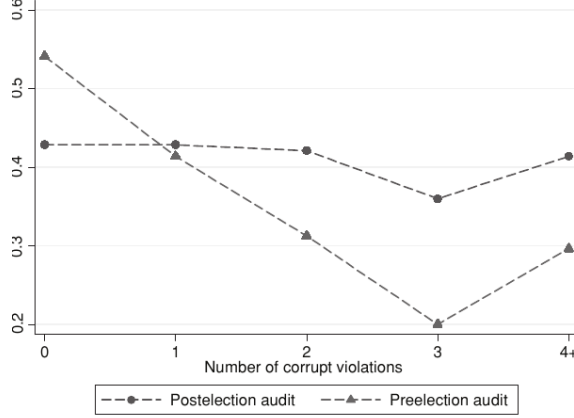


FIGURE III

Relationship between Reelection Rates and Corruption Levels

Notes. Figure shows the unadjusted relationship between the proportion of re-elected mayors who were reelected in the 2004 elections and the number of corruption incidents reported in the audit reports for municipalities audited before and after the elections. The points represented by circles are calculated for the municipalities that audited after the elections. The points represented by triangles are calculated for the municipalities audited before the elections. The figure was calculated for our entire sample of 373 municipalities based on data from Brazil's Electoral Commission and the CGU audit reports.

Figure III: reelection rates by corruption level

6.3 Tables III and IV: regression evidence across electoral outcomes

Read:

- Table III quantifies the relationship between audit release, corruption levels, and reelection probability.
- Linear and flexible models show that released information about corruption reduces reelection rates.
- Table IV shows similar patterns for vote share, win margin, and changes relative to the prior election.

Economic message: The result is not confined to a binary reelection outcome. Corruption information reduces broader measures of incumbent electoral performance.

TABLE III
THE EFFECTS OF THE RELEASE OF THE AUDITS ON REELECTION RATES BY THE LEVEL OF REPORTED CORRUPTION

	Linear		Quadratic	Semiparametric	Corruption ≤ 5	Corruption ≤ 4
	(1)	(2)	(3)	(4)	(5)	(6)
reelection audit	0.029 [0.083]	0.030 [0.082]	0.126 [0.101]	0.084 [0.104]	0.068 [0.087]	0.086 [0.088]
reelection audit × number of corrupt violations	-0.038 [0.035]	-0.038 [0.035]	-0.200 [0.090]*	-0.200 [0.090]*	-0.070 [0.041]+	-0.088 [0.043]*
reelection audit × number of corrupt violations ²			0.034 [0.017]*			
reelection audit × corruption = 0				0.010 [0.156]		0.003 [0.036]
reelection audit × corruption = 2				-0.253 [0.148]+		
reelection audit × corruption = 3				-0.321 [0.192]+		
reelection audit × corruption = 4+				-0.159 [0.168]		
number of corrupt violations	-0.013 [0.026]	-0.012 [0.027]	0.037 [0.066]		0.012 [0.033]	0.003 [0.036]
number of corrupt violations ²			-0.009 [0.011]			

Table III, part 1: reelection rates by corruption

TABLE III
(CONTINUED)

	Linear		Quadratic	Semiparametric	Corruption ≤ 5	Corruption ≤ 4
	(1)	(2)	(3)	(4)	(5)	(6)
				0.028 [0.126]		
				0.052 [0.114]		
				-0.006 [0.129]		
				-0.002 [0.136]		
	373	373	373	373	362	351
	0.05	0.18	0.19	0.22	0.19	0.20
			.089	.192		
cs	Yes	Yes	Yes	Yes	Yes	Yes
	No	Yes	Yes	Yes	Yes	Yes
	No	Yes	Yes	Yes	Yes	Yes

Notes: The effects of the release of the audits on the likelihood of reelection, by the level of corruption reported in the audits. Each column presents the results of an OLS regression of the dependent variable on an indicator for whether the mayor was reelected in the 2004. Except for column (1), all regressions include municipal characteristics: population (persons/km²), percentage of the population that is literate, percentage of the population that lives in the urban sector, per capita income expressed in logarithms, Gini index for income, effective number of political parties in the 2000 mayor election, municipal police (1/0), small claims court (1/0), judiciary district (1/0), municipal characteristics: 0 for male, age, married (1/0), education level, party dummies, and state intercepts. The estimation sample includes all mayors who were eligible for reelection. Robust standard errors are displayed in brackets. Significantly different from zero at 99 (**), 95 (*), and 10 (+) % confidence. In columns (3) and (4), the P-test tests the joint significance of the quadratic and cubic terms.

Table III, part 2: flexible corruption specifications

Dependent variables:	Pr(reelection)			Margin of victory			Vote share			Change in vote share		
	Full sample	Corruption ≤ 5	Semi-parametric	Full sample	Corruption ≤ 5	Semi-parametric	Full sample	Corruption ≤ 5	Semi-parametric	Full sample	Corruption ≤ 5	Semi-parametric
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
election audit	0.045 [0.005]	0.072 [0.099]	0.058 [0.135]	0.007 [0.037]	0.053 [0.039]	0.018 [0.053]	0.078 [0.102]	0.104 [0.106]	0.077 [0.146]	-0.014 [0.027]	0.006 [0.027]	0.012 [0.035]
election audit × corrupt violations	-0.06 [0.039]	-0.096 [0.046]		-0.034 [0.015]	-0.049 [0.019]		-0.078 [0.041]	-0.104 [0.048]	-0.01 [0.012]	-0.029 [0.013]		
election audit × corruption = 0			0.064 [0.188]			0.069 [0.071]			0.103 [0.201]		0.009 [0.046]	
election audit × corruption = 2			-0.335 [0.188]			-0.152 [0.079]			-0.42 [0.205]		-0.117 [0.054]	
election audit × corruption = 3			-0.321 [0.246]			-0.118 [0.079]			-0.371 [0.262]		-0.052 [0.062]	
election audit × corruption = 4+			-0.156 [0.195]			-0.082 [0.083]			-0.182 [0.208]		-0.045 [0.062]	
number of corrupt violations	-0.016 [0.030]	0.001 [0.036]		0.011 [0.012]	0.019 [0.014]		-0.002 [0.032]	0.014 [0.039]		-0.001 [0.010]	0.01 [0.010]	
election = 0			-0.006 [0.155]			0.011 [0.057]			-0.017 [0.166]			0.003 [0.035]

Table IV, part 1: vote share and margin outcomes

Dependent variables:	Pr(reelection)			Margin of victory			Vote share			Change in vote share		
	Full sample	Corruption ≤ 5	Semi-parametric	Full sample	Corruption ≤ 5	Semi-parametric	Full sample	Corruption ≤ 5	Semi-parametric	Full sample	Corruption ≤ 5	Semi-parametric
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
election audit												
election audit × corrupt violations												
election audit × corruption = 0												
election audit × corruption = 2												
election audit × corruption = 3												
election audit × corruption = 4+												
number of corrupt violations												
election = 0												

s the effects of the release of the audits on other electoral outcomes, by the level of corruption reported in the audits. Each column presents the results of an OLS regression where the dependent variable is listed in that column. All regressions include municipal characteristics: population density (persons/km), percentage of the population that lives in the urban sector, per capita income expressed in logarithms, Gini coefficient for income, effective number of mayor elections, municipal police (1/0), small claims court (1/0), judiciary district (1/0), mayoral characteristics: sex (1/0 for male), age, married (1/0), ties, and state intercepts. The estimation sample includes all mayors that ran for reelection and is listed in each column. Robust standard errors are in parentheses. * indicates different than zero at 90% (**), 95% (***), 99% (****) % confidence. In columns (3) and (4), the F-test tests the joint significance of the interaction terms.

Table IV, part 2: changes relative to 2000

6.4 Table V: audit quality and robustness checks

Read:

- Table V tests whether audit intensity, auditor behavior, political connections, or local characteristics explain the main estimates.
- The main audit-release-by-corruption effect remains stable across specifications.
- This reduces concerns that preselection audits were systematically different from postselection audits.

Economic message: The core result is unlikely to be an artifact of audit manipulation or differential audit quality.

Dependent variables:	Corruption		Pr(reelection)			
	Full sample	Corruption ≤ 5	Semiparametric	Corruption ≤ 5	Semiparametric	
	(1)	(2)	(3)	(4)	(5)	(6)
reelection audit	-0.332 [0.261]	-0.231 [0.298]	0.096 [0.125]	0.096 [0.138]	0.094 [0.129]	0.038 [0.162]
reelection audit × corrupt violations			-0.071 [0.039]		-0.081 [0.056]	
reelection audit × corruption = 0				-0.012 [0.155]		0.032 [0.185]
reelection audit × corruption = 2				-0.173 [0.146]		-0.14 [0.162]
reelection audit × corruption = 3				-0.364 [0.214]		-0.24 [0.314]
reelection audit × corruption = 4+				-0.153 [0.169]		-0.237 [0.213]
reelection audit × member of the governor's coalition	-0.155 [0.256]	-0.155 [0.388]	0.059 [0.134]	0.036 [0.136]		
reelection audit × margin of victory in 2000 elections			-0.638 [0.868]	-0.198 [0.316]		
reelection audit × PT	-0.004 [0.861]	-0.034 [0.864]	0.3 [0.278]	0.274 [0.264]		
reelection audit × PMDB	0.157 [0.389]	0.132 [0.398]	0.073 [0.130]	0.095 [0.129]		

Table V, part 1: audit quality and political controls

Dependent variables:	Corruption		Pr(reelection)			
	Full sample	Corruption ≤ 5	Semiparametric	Corruption ≤ 5	Semiparametric	
	(1)	(2)	(3)	(4)	(5)	(6)
reelection audit × PFL	0.064 [0.445]	0.052 [0.455]	-0.101 [0.149]		-0.146 [0.115]	
reelection audit × PSDB	-0.456 [0.989]	-0.471 [0.978]	-0.533 [0.241]		-0.263 [0.327]	
reelection audit × PSB	0.095 [0.628]	0.073 [0.637]	-0.46 [0.253]		-0.46 [0.259]	
reelection audit × PTB	-0.549 [0.591]	-0.562 [0.594]	0.232 [0.227]		0.313 [0.213]	
elections	373 0.35	373 0.35	362 0.27	373 0.29	246 0.22	255 0.22
other election terms (ues)	.97	.97	.08	.18		
corruption election terms (ues)				.34		.57

Columns (1) and (2) present the results of an OLS regression where the dependent variable is the number of violations associated with corruption. Columns (3)–(6) present the results of a semiparametric regression where the dependent variable is an indicator for whether an eligible incumbent was reelected in the 2004 elections. All regressions include municipal characteristics: population density (persons/km), percentage of the population that is literate, percentage of the population that lives in the urban sector, per capita income expressed in logarithms, Gini coefficient for income, margin of victory in the 2000 elections, an indicator for whether the mayor is a member of the governor's coalition, municipal police (1/0), small claims court (1/0), judiciary district (1/0), mayoral characteristics: sex (1/0 for male), age, married (1/0), education level, party dummies, and state intercepts. The regression in column (3) and (4) also includes the number of violations associated with corruption. The regression in column (5) and (6) also includes indicators for each local

Table V, part 2: robustness checks

6.5 Table VI and Figure IV: local radio and dissemination

Read:

- Table VI shows that local radio amplifies the effect of corruption revelations on incumbent reelection.
- When no corruption is revealed, local radio can increase reelection probability by spreading good news.
- Figure IV shows the nonparametric relationship among reelection, corruption, pre/post audit timing, and radio presence.

Economic message: Information disclosure and media access are complements. Audits matter most when voters hear about them.

Dependent variable: Pr(reelection)	Full sample (1)	Corruption ≤ 5 (2)	Demographic interactions (3)	Demographic and institutional interactions (4)	Households w/ radio (5)
reelection audit	-0.059 [0.091]	-0.033 [0.096]	0.296 [1.121]	0.208 [1.247]	-0.954 [0.629]
umber of corrupt violations	-0.034 [0.029]	-0.013 [0.035]	-0.13 [0.224]	-0.069 [0.288]	-0.161 [0.194]
umber of radio stations	-0.131 [0.064]*	-0.150 [0.063]*	-0.216 [0.073]**	-0.253 [0.083]**	
reelection audit $\times$ number of radio stations	0.229 [0.099]*	0.271 [0.104]**	0.356 [0.115]**	0.449 [0.129]**	
reelection audit $\times$ number of corrupt violations	0.007 [0.038]	-0.018 [0.044]	-0.236 [0.402]	-0.412 [0.430]	0.458 [0.229]*
umber of corrupt violations $\times$ number of radio stations	0.050 [0.026]+	0.058 [0.025]*	0.082 [0.025]**	0.09 [0.028]**	
reelection audit $\times$ corrupt violations $\times$ radio stations	-0.118 [0.045]**	-0.157 [0.067]*	-0.185 [0.051]**	-0.238 [0.064]**	

variable: Pr(reelection)	Full sample (1)	Corruption ≤ 5 (2)	Demographic interactions (3)	Demographic and institutional interactions (4)	Households w/ radio (5)
households with radio					-0.834 [0.782]
audit $\times$ households w/ radio					1.225 [0.752]
corrupt violations $\times$ audit $\times$ households w/ radio					0.181 [0.243]
audit $\times$ households w/ radio					-0.645 [0.292]*
is	373	362	373	373	373
ic interactions	No	No	Yes	Yes	No
il interactions	No	No	No	Yes	No

table reports in each column the result of an OLS linear probability model where the dependent variable is an indicator for whether the mayor was reelected in the 2004 election. The sample in all columns includes all mayors that were eligible for reelection. Number of radio stations is the number of local AM radio stations in a municipality. Households with radio is the total number of households that own at least one radio divided by the total number of households in the municipality. All regressions include municipal characteristics: population density (persons/km), percentage of the population that is literate, percentage of the population that lives in urban areas, per capita income expressed in logarithms, Gini coefficient for income, effective number of political parties in the 2000 mayor election, municipal police (1/10), small claims court (1/10), district judge (1/10), municipal characteristics: sex (1/10 for male), age, married (1/10), education level, party dummies, and state intercepts. Demographic interactions in column (4) are constructed by multiplying each one of the demographic variables by the reelection audit indicator and the number of corrupt violations. Demographic variables include: population density (persons/km), percentage of the population that is literate, percentage of the population that lives in urban areas, per capita income expressed in logarithms, for income. All lower term interactions are also included in the regressions. Institutional interactions in column (4) are constructed by multiplying each one of the variables by the reelection audit indicator and the number of corrupt violations. Institutional variables include presence of a local judge (1/10), effective number of parties or election, and presence of a small claims court (1/10). Robust standard errors are displayed in brackets. Significantly different from zero at 99 (**), 95 (*), 90 (+) %

Table VI, part 1: radio and corruption interactions

Table VI, part 2: alternative media specifications

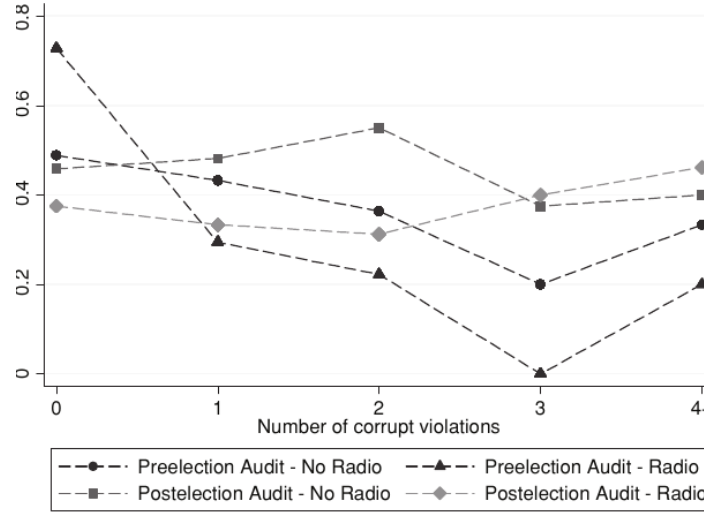


FIGURE IV
Relationship between Reelection Rates and Corruption Levels

Notes. Figure shows the unadjusted relationship between the proportion of reelected mayors who were reelected in the 2004 elections and the number of corrupt incidents reported in the audit reports for municipalities audited before and after the elections and the existence of local radio. The points represented by circles are calculated for the municipalities that were audited before the elections and do not have a local AM radio station. The points represented by triangles are calculated for the municipalities that were audited before the elections and have a local AM radio station. The points represented by squares are calculated for the municipalities that were audited after the elections and do not have a local AM radio station. The points represented by diamonds are calculated for municipalities that were audited after the elections and had a local AM radio station. The figure was calculated for our entire sample of 373 municipalities based on data from Brazil's Electoral Commission, the CGU audit reports, and IBGE.

Figure IV: corruption and reelection by radio presence

7 Short summary

- The paper uses randomized timing of Brazilian municipal audits to estimate how corruption information affects elections.
- The main treatment is whether audit results were publicly released before the 2004 election.
- The control group contains municipalities audited after the election.
- Audit release has little average effect unless the corruption content of the report is considered.
- Voters punish incumbents when audits reveal multiple corrupt violations.

- The punishment is stronger in municipalities with local radio stations.
- Radio also rewards clean incumbents by disseminating favorable audit information.
- The results are robust to checks for audit manipulation, political connections, and correlates of corruption and media access.
- The paper demonstrates that credible public information improves electoral accountability and that media determine how strongly information affects voter behavior.

8 Conclusion

8.1 Core conclusion

Ferraz and Finan show that voters punish corrupt politicians when credible information about corruption is released before elections. The effect is not simply anti-incumbency; it depends on the revealed level of corruption and is stronger when local radio can disseminate the information.

8.2 Economic intuition

Elections discipline politicians only when voters know enough about incumbent behavior. Random audits create new credible information. Voters update their beliefs and adjust their votes. Media increase the reach of the information, making the electoral punishment or reward stronger.

8.3 Takeaway

The paper provides clean causal evidence that transparency and media access improve democratic accountability. Disclosure policies can change political selection, but their impact depends on whether voters receive and understand the information.